

Highlights

Stablecoin Shocks, Spillovers and State-Dependent Transmission: Structural Identification and Time-Varying Propagation

稳定币冲击、溢出与状态依赖传导：结构识别与时变传播

Author Name

- Three identification strategies (heteroskedasticity, SVAR-IV, max-share) cross-validate.
- 三种识别策略（异方差识别、SVAR-IV、max-share）相互交叉验证。
- A 1% stablecoin shock lowers the 1M Treasury yield by up to 7.5–7.6 bps at a trough near 26 weeks.
- 1% 的稳定币冲击使 1 个月期国债收益率在约 26 周处累计最多下行 7.5–7.6 个基点。
- Stablecoin spillovers are neutral on average but turn net-positive around crisis episodes.
- 稳定币溢出在全样本上近似中性，但在危机时段转为净输出。
- Post-2023, high-rate and post-regulation states amplify the yield response 1.3–2.6 $\times$.
- 2023 年之后、高利率与后监管状态将收益率响应放大约 1.3–2.6 倍。
- Regulatory paradox: safer reserves tighten the stablecoin–bill market coupling.
- 监管悖论：更安全的储备结构反而强化了稳定币与国库券市场的耦合。

Stablecoin Shocks, Spillovers and State-Dependent
Transmission:
Structural Identification and Time-Varying Propagation
稳定币冲击、溢出与状态依赖传导：结构识别与时变传播

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Abstract

We identify structural stablecoin demand shocks and trace their dynamic transmission to the U.S. Treasury market, the dollar, crypto assets and equities, with particular attention to time variation and state dependence. Using a manually curated event library of 16 stablecoin-ecosystem news items (2021–2025) and daily market-cap changes of USDT and USDC over 2020-01–2026-06 (1,257 trading days), we deploy three identification strategies that cross-validate each other: (i) a Rigobon (2003) heteroskedasticity-based estimator applied to event vs. control dates; (ii) a daily SVAR identified by an event-based external instrument (SVAR-IV); and (iii) a max-share identification that selects the structural shock maximizing the stablecoin’s own forecast-error variance. The heteroskedasticity premise is supported (event-window variance ratio 1.90, Brown–Forsythe $p = 0.045$), but event-study point estimates are imprecise given the small event sample. The SVAR-IV (first-stage $F = 19.0$) and max-share identifications deliver a consistent core result: a stablecoin demand shock that lifts combined USDT+USDC market cap by 1% lowers the 1-month Treasury yield by up to 7.5–7.6 bps cumulatively at a trough near 26 weeks, and depreciates the broad dollar by 0.07–0.25%. Diebold–Yilmaz and time-varying spillover analyses show the stablecoin market is approximately neutral on average (net spillover -0.2 pp) but turns into a net transmitter around crisis episodes such as the Terra collapse and in late 2023–early 2024. State-dependent local projections reveal the

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yield response is $1.3\text{--}2.6\times$ larger in post-2023, high-rate and post-regulation states, and quantile local projections show the response is concentrated in the tails. A balance-sheet counterfactual calibrated to the current \$185 bn USDT market implies that a 10% USDT redemption would raise short yields by roughly 52 bps under a sister-study benchmark, 82 bps under our symmetric calibration, and up to 344 bps under an asymmetric outflow calibration, subject to linear-extrapolation caveats. The results expose a regulatory paradox: forcing reserves into short Treasuries improves safety but tightens the structural coupling between stablecoins and the bill market.

本文识别结构性稳定币需求冲击，并追踪其对美国国债市场、美元、加密资产与股票市场的动态传导，重点关注传导的时变性与状态依赖性。基于人工整理的 16 个稳定币生态新闻事件库（2021–2025 年）以及 2020 年 1 月至 2026 年 6 月（共 1,257 个交易日）USDT 与 USDC 的日度市值变动数据，本文采用三种相互交叉验证的识别策略：(i) 应用于事件日与对照日的 Rigobon (2003) 异方差识别估计量；(ii) 以事件日为外部工具变量的日度结构向量自回归 (SVAR-IV)；(iii) 选取最大化稳定币自身预测误差方差份额的结构冲击的 max-share 识别。异方差前提得到数据支持（事件窗口方差比 1.90，Brown–Forsythe 检验 $p = 0.045$ ），但受事件样本量限制，事件研究点估计不够精确。SVAR-IV（第一阶段 $F = 19.0$ ）与 max-share 识别给出一致的核心结论：使 USDT+USDC 合计市值上升 1% 的稳定币需求冲击，会在约 26 周处使 1 个月期国债收益率累计最多下行 7.5–7.6 个基点，并使广义美元指数贬值 0.07–0.25%。Diebold–Yilmaz 与时变溢出分析表明，稳定币市场在全样本上近似中性（净溢出 -0.2 个百分点），但在 Terra 崩盘等危机时段以及 2023 年末至 2024 年初转为净输出者。状态依赖局部投影显示，在 2023 年之后、高利率与后监管状态下，收益率响应放大约 1.3–2.6 倍；分位数局部投影进一步表明响应集中在分布尾部。以当前 1,850 亿美元的 USDT 市场规模校准的资产负债表反事实显示，10% 的 USDT 赎回在姊妹篇基准下将推高短期收益率约 52 个基点，在本文对称校准下约 82 个基点，在非对称流出校准下上限约 344 个基点，但需注意线性外推的局限。研究结果揭示了一个监管悖论：强制将储备集中于短期国债虽提升了个体安全性，却强化了稳定币与国库券市场之间的结构性耦合。

Keywords: Stablecoins, Treasury market, SVAR, Heteroskedasticity identification, Spillovers, State dependence, 稳定币, 国债市场, 溢出效应, 状态依赖, *JEL classification:* E44, G12, G15, G23

1. Introduction / 引言

Stablecoins have evolved from a niche settlement token into a structural shock transmitter for U.S. dollar financial markets. With combined reserves exceeding \$150 bn of short-dated Treasuries, the largest issuers—Tether (USDT) and Circle (USDC)—now function as marginal buyers whose demand and redemption behavior propagates through the bill market, the dollar, the crypto complex and, to a lesser extent, equities. While a companion paper [1] documents the average causal effect of stablecoin flows on bill yields, three questions remain open and motivate this paper. *First*, can a structural stablecoin shock—rather than a reduced-form flow—be identified from the data, and does it transmit dynamically across asset classes? *Second*, is the transmission stable over time, or does it intensify in stress, in bear markets, and after regulation? *Third*, what does the answer imply for the design of stablecoin reserve regulation?

稳定币已从一种小众的结算代币演变为美元金融市场的结构性冲击传导者。最大的发行方——Tether (USDT) 与 Circle (USDC)——合计持有超过 1,500 亿美元的短期国债储备, 已成为边际买家, 其需求与赎回行为会经由国库券市场、美元、加密资产体系乃至 (在较小程度上) 股票市场向外传导。姊妹篇论文 [1] 记录了稳定币流量对国库券收益率的平均因果效应, 但仍有三个问题悬而未决, 构成本文的研究动机。第一, 能否从数据中识别出结构性稳定币冲击 (而非约简形式的流量), 并追踪其跨资产类别的动态传导? 第二, 传导在时间上是否稳定, 还是会在压力时期、熊市以及监管落地之后强化? 第三, 上述答案对稳定币储备监管的设计意味着什么?

Identification is the central challenge. Stablecoin market-cap changes are confounded by simultaneous feedback from financial markets, by common macro-finance factors, and by broad crypto-market events. A naive regression of yields on stablecoin growth conflates these channels. We deploy three identification strategies that exploit different but complementary sources of variation and serve as mutual cross-validation.

识别是核心挑战。稳定币市值变动同时受到金融市场反向反馈、共同宏观金融因子以及广义加密市场事件的混杂影响; 对收益率做稳定币增长率的朴素回归会混淆这些渠道。本文部署三种识别策略, 它们利用不同但互补的变异来源, 并互为交叉验证。

1. **Heteroskedasticity identification** [2]. We construct a manually curated event library of 16 stablecoin-ecosystem news items over 2021–2025, retaining only events that originate within the stablecoin ecosystem and are not triggered by macro or broad-crypto news. On event

dates the variance of the stablecoin-specific shock is elevated while the variance of other shocks is unchanged; the resulting covariance structure identifies the causal effect of stablecoin growth on financial variables.

2. 异方差识别 [2]。我们人工整理了 2021–2025 年间 16 个稳定币生态新闻事件，仅保留源自稳定币生态内部、且非由宏观或广义加密新闻触发的事件。在事件日，稳定币特有冲击的方差上升，而其他冲击的方差不变；由此产生的协方差结构差异即可识别稳定币增长对金融变量的因果效应。
3. **SVAR–IV**. We estimate a daily VAR in stablecoin market-cap growth, the 1M Treasury yield change, Bitcoin returns, S&P 500 returns and broad-dollar returns, and identify the stablecoin structural shock using event-window market-cap growth as an external instrument in the spirit of Mertens–Ravn.
4. **SVAR–IV**。我们估计一个包含稳定币市值增长率、1 个月期国债收益率变动、比特币收益率、标普 500 收益率与广义美元收益率的日度 VAR，并以事件窗口内的市值增长作为外部工具变量（沿袭 Mertens–Ravn 的思路）识别稳定币结构冲击。
5. **Max-share identification** [3]. Without relying on narrative events, we select the structural shock that maximizes the stablecoin’s own forecast-error variance share over a finite horizon, subject to orthogonality with the dominant shocks of the other variables.
6. **Max-share** 识别 [3]。不依赖叙事性事件，我们选取在有限 horizon 内最大化稳定币自身预测误差方差份额的结构冲击，并要求其与其余变量的主导冲击正交。

The three strategies deliver a consistent core message, albeit with different precision. The heteroskedasticity premise holds in the data: three-day event windows exhibit 1.90 times the variance of matched control windows (Brown–Forsythe $p = 0.045$). Event-study point estimates have the expected signs for short yields, the dollar and crypto but are imprecise, reflecting the limited power of a 16-event sample (first-stage $F = 1.6$). The SVAR–IV, by contrast, exploits the full daily panel: the external instrument is strong (first-stage $F = 19.0$), and a 1% stablecoin demand shock lowers the 1M Treasury yield cumulatively by up to 7.6 bps at a trough around 130 trading days (about 26 weeks), with an impact effect of -0.24 bp. The max-share identification—which uses no narrative information—reproduces the yield trough (-7.5 bps at 26 weeks) and the dollar depreciation almost exactly,

supporting robustness of the structural identification. The cross-sectional equity analysis planned along the lines of (author?) [4] could not be implemented because daily equity data for Coinbase, PayPal, Block and the bank indices are not available in our current data sources; we leave it to future work.

三种策略传递了口径一致的核心信息，尽管精度不同。异方差前提在数据中成立：三日事件窗口的方差是匹配对照窗口的 1.90 倍（Brown–Forsythe 检验 $p = 0.045$ ）。事件研究点估计在短期收益率、美元与加密资产上符号符合预期，但因 16 个事件的样本功效有限而不够精确（第一阶段 $F = 1.6$ ）。相比之下，SVAR-IV 利用了完整的日度面板：外部工具变量较强（第一阶段 $F = 19.0$ ），1% 的稳定币需求冲击使 1 个月期国债收益率在约 130 个交易日（约 26 周）处累计最多下行 7.6 个基点，冲击当期效应为 -0.24 个基点。完全不使用叙事信息的 max-share 识别几乎分毫不差地复现了收益率谷底（26 周处 -7.5 个基点）与美元贬值，支持结构识别的稳健性。原计划参照 (author?) [4] 开展的跨股票横截面分析，因当前数据源不包含 Coinbase、PayPal、Block 及银行板块指数的日度行情而无法实施，留待后续研究。

We then study time variation. A Diebold–Yilmaz [5] spillover decomposition over the full daily sample shows the stablecoin market is approximately neutral on average (net spillover -0.17 percentage points), so H2 in its strong, full-sample form is not supported. However, a time-varying rolling-window analysis [6] reveals that net transmission from stablecoins turns positive around the Terra collapse (May 2022, window mean $+1.1$ pp) and reaches its peak in October 2023–February 2024 (up to $+5.6$ pp), while remaining negative in calm periods: transmission is state-dependent rather than constant.

随后我们研究时变性。基于全日度样本的 Diebold–Yilmaz [5] 溢出分解显示，稳定币市场平均而言近似中性（净溢出 -0.17 个百分点），因此强形式的 H2（全样本净输出）未获支持。但基于滚动窗口的时变分析 [6] 表明，稳定币的净传导在 Terra 崩盘（2022 年 5 月，窗口均值 $+1.1$ 个百分点）前后转正，并在 2023 年 10 月至 2024 年 2 月达到峰值（最高 $+5.6$ 个百分点），而在平静期为负：传导是状态依赖的，而非常数。

State-dependent local projections confirm this pattern for levels as well: the peak yield response to a 1% stablecoin shock is -11.5 bps at 24 trading days in the baseline, and is $2.4\times$ larger after 2023, $2.2\times$ larger in high-rate states, and $2.6\times$ larger after the GENIUS Act, consistent with a regulatory-paradox interpretation: requiring reserves in short Treasuries raises safety but deepens the structural coupling. Quantile local projections show the

response is essentially zero at the median but reaches -29 bps at the 0.9 quantile, i.e., stablecoin shocks matter disproportionately in the tails of the yield-change distribution.

状态依赖局部投影在水平值层面证实了这一模式：基准情形下 1% 稳定币冲击的收益率峰值响应为 24 个交易日处的 -11.5 个基点；2023 年之后放大 2.4 倍，高利率状态放大 2.2 倍，GENIUS 法案之后放大 2.6 倍——这与监管悖论的解释一致：强制储备集中于短期国债提升了安全性，却加深了结构性耦合。分位数局部投影显示，中位数处响应接近于零，而 0.9 分位处高达 -29 个基点，即稳定币冲击在收益率变动分布的尾部具有不成比例的重要性。

Finally, we run a balance-sheet counterfactual calibrated to the current USDT market (\$185 bn end of sample): a 10% USDT redemption implies roughly \$11 bn of forced bill liquidation. Translating this sell pressure through three calibrations—the sister-study benchmark [1], our symmetric LP coefficient, and our asymmetric outflow coefficient—yields a short-yield impact of 52 bps, 82 bps and up to 344 bps respectively, with the last figure best read as an upper bound under linear extrapolation.

最后，我们以当前 USDT 市场规模（样本期末 1,850 亿美元）校准资产负债表反事实：10% 的 USDT 赎回意味着约 110 亿美元的被迫国库券抛售。将这一抛售压力分别通过三种校准——姊妹篇基准 [1]、本文对称 LP 系数、本文非对称流出系数——转化为收益率影响，得到短期收益率上行 52 个基点、82 个基点与上限约 344 个基点；其中最后一个数字应理解为线性外推下的上界。

Contribution.. Relative to (author?) [4], who also deploy narrative events and SVAR-IV, we (i) add the max-share cross-validation, (ii) integrate Diebold-Yilmaz and time-varying spillover measurement, (iii) systematically characterize state dependence along five dimensions (early/late, bull/bear, vol regime, rate regime, pre/post regulation), and (iv) embed the estimates in a balance-sheet fire-sale counterfactual with explicit regulatory-paradox implications. Relative to (author?) [1], we move from reduced-form flow effects to structural shock identification and emphasize time variation rather than average effects.

研究贡献。相对于同样采用叙事事件与 SVAR-IV 的 (author?) [4]，本文 (i) 增加了 max-share 交叉验证；(ii) 整合了 Diebold-Yilmaz 与时变溢出度量；(iii) 沿五个维度（早期/晚期、牛市/熊市、波动率区制、利率区制、监管前/后）系统刻画了状态依赖性；(iv) 将估计结果嵌入带有明确监

管悖论含义的资产负债表挤兑反事实。相对于 (author?) [1], 本文从约简形式的流量效应推进到结构冲击识别, 并强调时变性而非平均效应。

The rest of the paper is organized as follows. Section 2 reviews the literature. Section 3 develops mechanisms and hypotheses. Section 4 describes data and the event library. Section 5 presents the three identification strategies. Section 6 reports dynamic IRFs. Section 7 studies volatility spillovers. Section 8 reports state-dependent results. Section 9 runs the balance-sheet counterfactual. Section 10 summarizes robustness checks. Section 11 concludes.

本文其余部分安排如下。第 2 节回顾文献; 第 3 节阐述机制与假设; 第 4 节介绍数据与事件库; 第 5 节给出三种识别策略; 第 6 节报告动态脉冲响应; 第 7 节研究波动溢出; 第 8 节报告状态依赖结果; 第 9 节进行资产负债表反事实分析; 第 10 节汇总稳健性检验; 第 11 节总结全文。

2. Related literature / 相关文献

This paper contributes to three literatures. First, the structural identification literature in macro-finance: (author?) [2] develops identification through heteroskedasticity, applied by (author?) [7] to monetary policy; (author?) [3] proposes max-share identification. We adapt both to the stablecoin context. Second, the spillover literature: (author?) [5] develop the variance-decomposition spillover index; (author?) [6] extend it to time-varying parameters. We apply both to stablecoin–bill–dollar–crypto linkages. Third, the stablecoin literature: (author?) [4] identify stablecoin shocks narratively; (author?) [1] estimate flow effects via LP–IV; (author?) [8] provides balance-sheet counterfactuals; (author?) [9] pioneer on-chain flow identification; and the preferred-habitat framework of (author?) [10] together with the safe-asset demand estimates of (author?) [11] supply the theoretical backdrop for bill-market price impact. We integrate these strands into a unified structural and state-dependent framework.

本文对三支文献有所贡献。第一支是宏观金融中的结构识别文献: (author?) [2] 提出经由异方差的识别方法, 并由 (author?) [7] 应用于货币政策; (author?) [3] 提出 max-share 识别。本文将二者适配到稳定币情境。第二支是溢出文献: (author?) [5] 提出基于方差分解的溢出指数; (author?) [6] 将其推广到时变参数框架。本文将二者应用于稳定币—国库券—美元—加密资产的联动。第三支是稳定币文献: (author?) [4] 以叙事方法识别稳定币冲击; (author?) [1] 用 LP–IV 估计流量效应; (author?) [8] 给出资产负债表反事实; (author?) [9] 开创了链上流量识别; (author?) [10] 的

偏好栖息地框架与 (author?) [11] 的安全资产需求估计则为国库券市场的价格冲击提供了理论背景。本文将这些线索整合进一个统一的、状态依赖的结构框架。

3. Theoretical mechanism and hypotheses / 理论机制与研究假设

3.1. From stablecoin demand to Treasury yields / 从稳定币需求到国债收益率

The transmission chain is:

stablecoin demand $\uparrow \Rightarrow$ issuer reserve growth $\Rightarrow$ short Treasury demand $\uparrow \Rightarrow y^{1M}, y^{3M} \downarrow$.

传导链条为：稳定币需求上升 $\Rightarrow$ 发行方储备增长 $\Rightarrow$ 短期国债需求上升 $\Rightarrow$ 1 个月与 3 个月期收益率下行。

Because reserves are concentrated in bills maturing under three months, the effect is localized at the short end. The issuer balance sheet reads $A_t^i = \text{TBill}_t^i + \text{Repo}_t^i + \text{Cash}_t^i + \text{Other}_t^i$, $L_t^i = SC_t^i$, so a 1% stablecoin demand shock translates into proportional Treasury demand growth at rate θ^i (the reserve share allocated to bills). In a preferred-habitat market [10], such incremental demand from a price-insensitive buyer compresses short-end yields.

由于储备集中于三个月以内到期的国库券，该效应局域于收益率曲线短端。发行方资产负债表可写为 $A_t^i = \text{TBill}_t^i + \text{Repo}_t^i + \text{Cash}_t^i + \text{Other}_t^i$, $L_t^i = SC_t^i$, 因此 1% 的稳定币需求冲击会按 θ^i （储备配置于国库券的比例）转化为同比例的国债需求增长。在偏好栖息地市场 [10] 中，来自价格不敏感买家的此类增量需求会压缩短端收益率。

3.2. Cross-asset propagation / 跨资产传播

Beyond bills, the shock propagates through portfolio rebalancing. Lower short yields reduce the attractiveness of dollar short-term assets, leading to dollar depreciation; easier on-chain dollar liquidity lowers frictions in crypto trading; and crypto sentiment may spill into equities, especially firms with stablecoin-infrastructure exposure.

除国库券外，冲击还通过组合再平衡向外传播：短端收益率下行降低了美元短期资产的吸引力，导致美元贬值；链上美元流动性宽松降低了加密交易的摩擦；加密市场情绪亦可能外溢至股票市场，尤其是具有稳定币基础设施敞口的公司。

3.3. Time variation and state dependence / 时变性与状态依赖

We hypothesize that the transmission has strengthened over time as stablecoin reserves grew, and that it amplifies in stress. Formally, let $\beta_h(s_t)$ denote the horizon- h response conditional on a state indicator s_t . We expect

$$|\beta_h(\text{late})| > |\beta_h(\text{early})|, \quad |\beta_h(\text{bear})| > |\beta_h(\text{bull})|, \quad |\beta_h(\text{stress})| > |\beta_h(\text{normal})|, \quad |\beta_h(\text{post-reg})| > |\beta_h(\text{pre-reg})| \quad (1)$$

我们假设：随着稳定币储备规模扩张，传导强度随时间增强，并在压力时期进一步放大。正式地，令 $\beta_h(s_t)$ 表示以状态指示变量 s_t 为条件的 horizon- h 响应，我们预期式 (1) 成立：晚期响应大于早期、熊市大于牛市、压力期大于正常期、监管后大于监管前。

3.4. Regulatory paradox / 监管悖论

Requiring reserves in short Treasuries (as under the U.S. GENIUS Act) raises safety but tightens the stablecoin–bill coupling, so post-regulation transmission may *strengthen* even as individual-issuer risk falls.

强制储备集中于短期国债（如美国 GENIUS 法案所要求）虽提升了个体发行方的安全性，却也收紧了稳定币与国库券市场的耦合，因此即便单个发行方风险下降，监管后的传导仍可能增强。

Table 1: 研究假设与实证含义 / Hypotheses and empirical implications.

	Hypothesis	Implication
H1	Structural shock transmits to yields, dollar, crypto	IRF sign pattern
H2	Stablecoin is net volatility transmitter	DY/TVP net spillover > 0
H3	Transmission intensifies in stress/bear/post-reg	$ \beta $ ratios > 1
H4	Regulatory paradox	post-reg coupling stronger

4. Data and event library / 数据与事件库

4.1. Market and macro data / 市场与宏观数据

We use daily data over 2020-01-01 to 2026-06-30 (1,257 trading days after merging). Treasury yields (1M, 3M, 1Y, 2Y, 10Y) are from FRED (DGS1MO, DTB3, DGS1, DGS2, DGS10). The broad dollar index (DTWEXBGS) and VIX (VIXCLS) are also from FRED [12]. Bitcoin price, S&P 500 index, gold price, DeFi total value locked (TVL) and the SOFR–RRP spread are from a merged macro–crypto daily panel. The SOFR–RRP spread serves as

our money-market stress proxy in the spillover system, because the BAML commercial-paper spread available to us starts only in June 2023 and the TED rate has been discontinued; relative to the event-study design of (author?) [4], we therefore also replace the Bloomberg Galaxy Crypto Index (BGCI) by Bitcoin returns and the MOVE index by the VIX.

本文使用 2020-01-01 至 2026-06-30 的日度数据（合并后共 1,257 个交易日）。国债收益率（1 个月、3 个月、1 年、2 年、10 年）取自 FRED（DGS1MO、DTB3、DGS1、DGS2、DGS10）；广义美元指数（DTWEXBGS）与 VIX（VIXCLS）同样来自 FRED [12]。比特币价格、标普 500 指数、黄金价格、DeFi 总锁仓量（TVL）以及 SOFR-RRP 利差来自一份合并的宏观—加密日度面板。SOFR-RRP 利差在溢出系统中充当货币市场压力代理，因为我们可获得的 BAML 商业票据利差 2023 年 6 月才开始，而 TED 利差已停更；同样出于数据可得性，相对 (author?) [4] 的事件研究设计，我们以比特币收益率替代彭博银河加密指数（BGCI），以 VIX 替代 MOVE 指数。

4.2. Stablecoin variables / 稳定币变量

We use daily market-cap (circulating supply) data for USDT and USDC, with DAI and BUSD available for robustness, and define the stablecoin growth rate as

$$\Delta m_t = m_t - m_{t-1}, \quad m_t = \log(SC_t^{\text{USDT}} + SC_t^{\text{USDC}}), \quad (2)$$

本文使用 USDT 与 USDC 的日度市值（流通供应量）数据，DAI 与 BUSD 供稳健性使用，并定义稳定币增长率为式 (2)：其中 SC_t^{USDT} 与 SC_t^{USDC} 分别为 USDT 与 USDC 在第 t 日的流通市值。where SC_t^{USDT} and SC_t^{USDC} denote the circulating market capitalization of USDT and USDC on day t .

The raw daily growth rate Δm_t has mean 0.21% and standard deviation 1.12%, with a maximum of +34.4% driven by a supply-series revision jump in April 2020 that lies outside every event and control window. For VAR-based exercises we winsorize Δm_t at the 0.5%/99.5% quantiles (Δm_t^w) to protect the lag-polynomial estimates from this data revision; the event-study and Rigobon exercises use the raw series, because winsorization would flatten exactly the large genuine shocks (UST, SVB) on which the heteroskedasticity identification relies.

原始日增长率 Δm_t 均值为 0.21%，标准差为 1.12%，最大值 +34.4% 来自 2020 年 4 月供应序列的一次修订跳点，该日不在任何事件或对照窗口内。在基于 VAR 的估计中，我们对 Δm_t 做 0.5%/99.5% 分位缩尾（记

为 Δm_t^w), 以免该数据修订污染滞后多项式估计; 而事件研究与 Rigobon 估计使用原始序列, 因为缩尾恰恰会压平异方差识别所依赖的真实大冲击 (UST、SVB)。

4.3. Event library / 事件库

Following (author?) [4], we construct an event library from systematic news searches over 2021–2025 using the keywords “stablecoin”, “stablecoins”, “USD Coin”, “USDC”, “Tether”, “USDT”, and retain events that (i) originate within the stablecoin ecosystem, (ii) are not triggered by macro-financial news, (iii) are not broad-crypto-market events, and (iv) are dated precisely enough to map to a trading day. The retained 16 events (Table 2) include regulatory and legal actions (e.g., the NYAG–Tether settlement of 2021-02-23; the GENIUS Act legislative sequence of 2025), reserve-composition events (Tether’s commercial-paper elimination), crisis events (Terra/UST collapse, SVB/USDC de-peg) and market/product events (PayPal’s PYUSD launch, Circle’s IPO). For each event we take a three-day window starting at the event trading day, and draw one matched three-day control window within ± 2 months, at least five trading days away from any event window (random seed 43).

参照 (author?) [4], 我们基于 2021–2025 年间以 “stablecoin”、“stablecoins”、“USD Coin”、“USDC”、“Tether”、“USDT” 为关键词的系统性新闻检索构建事件库, 仅保留满足以下条件的事件: (i) 源自稳定币生态内部; (ii) 非由宏观金融新闻触发; (iii) 非广义加密市场事件; (iv) 日期足够精确以映射到交易日。保留的 16 个事件 (表 2) 涵盖监管与法律行动 (如 2021-02-23 纽约总检察长与 Tether 的和解; 2025 年 GENIUS 法案立法序列)、储备构成事件 (Tether 清零商业票据)、危机事件 (Terra/UST 崩盘、SVB/USDC 脱锚) 以及市场/产品事件 (PayPal 推出 PYUSD、Circle 上市)。每个事件取自事件交易日起的三日窗口, 并在 ± 2 个月内、距任何事件窗口至少 5 个交易日处随机抽取一个匹配的三日对照窗口 (随机种子 43)。

Event-window observations exhibit a mean absolute market-cap change of 0.325% versus 0.213% in control windows ($n = 48$ observations each), and the event-window variance is 1.90 times the control variance (Brown–Forsythe statistic 4.12, $p = 0.045$), supporting the heteroskedasticity premise required by Eq. (5) below.

事件窗口观测的平均绝对市值变动为 0.325%, 对照窗口为 0.213% (各 48 个观测); 事件窗口方差为对照窗口的 1.90 倍 (Brown–Forsythe 统计量 4.12, $p = 0.045$), 支持下文式 (5) 所需的异方差前提。

Table 2: 稳定币事件库：16 个事件的三日窗口市值变动 / Event library: three-day market-cap changes around the 16 retained events.

Date	Event	Type	Δm event (%)	Δm control (%)
2021-02-23	NY AG–Tether settlement (\$18.5M fine)	Legal	1.80	2.74
2021-10-15	CFTC fines Tether \$41M over reserve claims	Legal	-0.48	0.60
2021-11-01	PWG report on stablecoin regulation	Regulatory	1.17	0.84
2022-05-09	Terra/UST collapse begins	Crisis	-0.87	0.67
2022-07-21	Tether commits to zero commercial paper	Reserves	-0.20	-0.13
2022-10-13	Tether completes CP elimination	Reserves	-0.96	-1.04
2023-02-13	NYDFS orders Paxos to halt BUSD issuance	Regulatory	1.43	-0.32
2023-03-10	SVB failure / USDC de-peg	Crisis	-1.14	-0.09
2023-08-07	PayPal launches PYUSD stablecoin	Product	0.00	-0.22
2024-07-01	MiCA stablecoin regime takes effect in EU	Regulatory	0.07	0.67
2024-12-11	Circle–Binance USDC partnership	Product	1.60	0.38
2025-02-04	GENIUS Act introduced in US Senate	Regulatory	1.57	0.06
2025-05-19	GENIUS Act advances in Senate cloture vote	Regulatory	0.77	0.32
2025-06-05	Circle IPO on NYSE	Product	0.18	0.47
2025-06-17	GENIUS Act passes US Senate	Regulatory	0.30	-0.01
2025-07-18	GENIUS Act signed into law	Regulatory	0.37	-0.12

注： Δm 为三日窗口内 USDT+USDC 合计市值的对数增长率（%）；对照窗口在 ± 2 个月内随机抽取（种子 43）。Notes: Δm is the three-day log growth of combined USDT+USDC market cap（%）; control windows are randomly drawn within ± 2 months (seed 43).

5. Identification / 识别策略

5.1. Heteroskedasticity-based event-study estimator / 基于异方差的事件研究估计量

Let Δm_t denote stablecoin market-cap growth and Δy_t an outcome variable. The simultaneous system is

$$\Delta m_t = \beta \Delta y_t + \gamma F_t + \nu_t, \quad (3)$$

$$\Delta y_t = \alpha \Delta m_t + F_t + \eta_t, \quad (4)$$

令 Δm_t 表示稳定币市值增长率, Δy_t 为结果变量。联立系统为式 (3)–(4): 其中 F_t 为不可观测的共同因子, ν_t 为稳定币特有冲击, η_t 为结果变量的自身冲击。where F_t is an unobserved common factor, ν_t the stablecoin shock, and η_t the own shock of the outcome.

Let S denote event dates and $\tilde{S}$ control dates. The identification assumption is

$$\sigma_{\nu,S}^2 > \sigma_{\nu,\tilde{S}}^2, \quad \sigma_{F,S}^2 = \sigma_{F,\tilde{S}}^2, \quad \sigma_{\eta,S}^2 = \sigma_{\eta,\tilde{S}}^2. \quad (5)$$

令 S 表示事件日集合， $\tilde{S}$ 为对照日集合。识别假设为式 (5)：稳定币冲击的方差在事件日更高，而共同因子与结果变量自身冲击的方差在两组之间不变。

Defining the covariance matrix Ω_j on sample $j \in \{S, \tilde{S}\}$, its difference satisfies

$$\Delta\Omega \equiv \Omega_S - \Omega_{\tilde{S}} = \frac{(\sigma_{\nu,S} - \sigma_{\nu,\tilde{S}})}{(1 - \alpha\beta)^2} \begin{bmatrix} \alpha^2 & \alpha \\ \alpha & 1 \end{bmatrix}, \quad (6)$$

yielding the closed-form estimator

$$\hat{\alpha} = \frac{\text{Cov}_S(\Delta y_t, \Delta m_t) - \text{Cov}_{\tilde{S}}(\Delta y_t, \Delta m_t)}{\text{Var}_S(\Delta m_t) - \text{Var}_{\tilde{S}}(\Delta m_t)}. \quad (7)$$

记样本 $j \in \{S, \tilde{S}\}$ 上的协方差矩阵为 Ω_j ，其差满足式 (6)，从而得到式 (7) 的闭式估计量：两组样本协方差之差除以两组稳定币增长率方差之差。

Operationally we implement Eq. (7) as the IV regression of Δy on Δm instrumented by $\Delta \tilde{m} \equiv (\Delta m'_S, -\Delta m'_{\tilde{S}})'$ [7], with heteroskedasticity-robust standard errors, and report the closed-form ratio alongside as a cross-check.

操作上，我们将式 (7) 实现为以 $\Delta \tilde{m} \equiv (\Delta m'_S, -\Delta m'_{\tilde{S}})'$ 为工具变量、对 Δy 关于 Δm 的 IV 回归 [7]，使用异方差稳健标准误，并同时报告闭式比值作为交叉验证。

5.2. SVAR-IV / 工具变量结构向量自回归

We estimate the daily reduced-form VAR

$$\mathbf{y}_t = \boldsymbol{\alpha} + \sum_{j=1}^p \mathbf{B}_j \mathbf{y}_{t-j} + \mathbf{u}_t, \quad \mathbf{y}_t = \begin{bmatrix} \Delta m_t^w \\ \Delta y_t^{1M} \\ \Delta \log \text{BTC}_t \\ \Delta \log \text{SP500}_t \\ \Delta \log \text{DXY}_t \end{bmatrix}, \quad (8)$$

我们估计式 (8) 的日度约简型 VAR：变量依次为缩尾后的稳定币市值增长率、1 个月期国债收益率变动、比特币对数收益、标普 500 对数收益与广义美元对数收益。

with $p = 13$ lags (about two and a half weeks of trading days). The structural mapping is $\mathbf{u}_t = \mathbf{S}\boldsymbol{\varepsilon}_t$, with the first column $\mathbf{s}_1$ the stablecoin shock's impact vector. The observed daily stablecoin growth is

$$\Delta m_t = \varepsilon_{1,t} + \sum_{i \neq 1} \varepsilon_{i,t} + \xi_t, \quad (9)$$

滞后阶数取 $p = 13$ (约两周半的交易日)。结构映射为 $\mathbf{u}_t = \mathbf{S}\boldsymbol{\varepsilon}_t$, 其第一列 $\mathbf{s}_1$ 即稳定币冲击的影响向量。观测到的日度稳定币增长满足式 (9): 其中 ξ_t 为测量误差。where ξ_t is measurement error.

The external instrument z_t equals event-window stablecoin growth on the 48 event dates and zero otherwise (a Mertens–Ravn-type narrative instrument). Under variance assumptions analogous to Eq. (5), the impact vector is identified by the cross-moment of the VAR residuals with the instrument, as in Eq. (10):

$$\hat{\mathbf{s}}_1 = \frac{\mathbf{U}'\Delta\mathbf{m}_S - \mathbf{U}'\Delta\mathbf{m}_{\tilde{S}}}{\Delta\mathbf{m}'_S\Delta\mathbf{m}_S - \Delta\mathbf{m}'_{\tilde{S}}\Delta\mathbf{m}_{\tilde{S}}}. \quad (10)$$

外部工具变量 z_t 在 48 个事件日取事件窗口稳定币增长值、其余日期取零 (Mertens–Ravn 式叙事工具变量)。在与式 (5) 类似的方差假设下, 影响向量由 VAR 残差与工具变量的交叉矩识别, 如式 (10), 其中 $\mathbf{U}$ 为 VAR 残差矩阵。

The first-stage heteroskedasticity-robust F -statistic is 19.0, above the weak-instrument threshold of 10. Inference uses a fixed-design moving block bootstrap with 2,000 replications (block length 13, matching the VAR lag length); 68% and 90% confidence bands are reported. The bands are wide—an honest reflection of weak-instrument uncertainty at long horizons—and we therefore emphasize the point-estimate agreement across identifications rather than band significance.

第一阶段异方差稳健 F 统计量为 19.0, 高于弱工具变量门槛 10。推断采用固定设计的移动区组自助法 (MBB), 重复 2,000 次 (区组长度 13, 与 VAR 滞后阶数一致), 报告 68% 与 90% 置信带。置信带较宽——这是长 horizon 上弱工具不确定性的如实反映——因此我们强调不同识别之间点估计的一致性, 而非置信带的显著性。

5.3. *Max-share identification* / *Max-share* 识别

For any impact matrix $\mathbf{S} = \tilde{\mathbf{S}}\mathbf{Q}$ with $\tilde{\mathbf{S}}$ the Cholesky factor of $\boldsymbol{\Sigma}$ and $\mathbf{Q}$ an orthonormal rotation, the share of variable i 's h -step forecast-error variance explained by structural shock j is

$$\mathcal{S}_{i,j}(h) = \frac{\sum_{\ell=0}^h \mathbf{B}_{i,\ell} \tilde{\mathbf{S}} \mathbf{q}_j \mathbf{q}_j' \tilde{\mathbf{S}}' \mathbf{B}_{i,\ell}'}{\sum_{\ell=0}^h \mathbf{B}_{i,\ell} \boldsymbol{\Sigma} \mathbf{B}_{i,\ell}'}. \quad (11)$$

对任意影响矩阵 $\mathbf{S} = \tilde{\mathbf{S}}\mathbf{Q}$ (其中 $\tilde{\mathbf{S}}$ 为 $\boldsymbol{\Sigma}$ 的 Cholesky 因子, $\mathbf{Q}$ 为正交旋转矩阵), 结构冲击 j 解释变量 i 的 h 步预测误差方差份额由式 (11) 给出。

For each non-stablecoin variable i we first obtain

$$\mathbf{q}_i^* = \arg \max_{\mathbf{q}_i} \sum_{h=0}^H \mathcal{S}_{i,i}(h) \quad \text{s.t.} \quad \mathbf{q}_i' \mathbf{q}_i = 1, \quad (12)$$

then identify the stablecoin shock as

$$\mathbf{q}_1^* = \arg \max_{\mathbf{q}_1} \sum_{h=0}^H \mathcal{S}_{1,1}(h) \quad \text{s.t.} \quad \mathbf{q}_1' \mathbf{q}_1 = 1, \quad \mathbf{q}_1' \mathbf{q}_i^* = 0 \quad \forall i \in \mathcal{J}_{-1}, \quad (13)$$

with $H = 20$ trading days. The structural impact vector is $\mathbf{s}_1 = \tilde{\mathbf{S}} \mathbf{q}_1^*$. Because this identification uses no narrative information, agreement with the SVAR-IV responses is a strong cross-validation of the event library.

我们先对每个非稳定币变量求解式 (12) 得到其主导冲击方向，再按式 (13) 识别稳定币冲击：在与所有其他主导方向正交的约束下，最大化稳定币自身预测误差方差份额。horizon 取 $H = 20$ 个交易日，结构影响向量为 $\mathbf{s}_1 = \tilde{\mathbf{S}} \mathbf{q}_1^*$ 。由于该识别完全不使用叙事信息，其与 SVAR-IV 响应的一致性 是事件库的强有力交叉验证。

6. Dynamic responses / 动态响应

6.1. Event-study estimates / 事件研究估计

Table 3 reports the heteroskedasticity-based event-study estimates of α (the effect of a 1% stablecoin market-cap increase), and Figure 1 visualizes them. Point estimates have the expected negative sign for the 3M yield (-1.80 bps, $t = -1.00$), the broad dollar (-0.33% , $t = -1.07$), Bitcoin (-6.60% , $t = -1.11$) and the S&P 500 (-1.22%), but none is statistically significant, and the first-stage F -statistic of 1.6 falls well below conventional thresholds. The closed-form Rigobon ratios (reported in the online appendix CSV) tell the same story. We read this as a power statement rather than a falsification: with only 16 events and 48 event-window observations, the heteroskedasticity contrast cannot pin down α precisely, even though the variance premise itself holds (Table 2). This motivates the panel-based SVAR-IV, which reuses the same events as an instrument within the full 1,257-day sample.

表 3 报告了基于异方差的事件研究估计 α (1% 稳定币市值上升的效应)，图 1 将其可视化。点估计在 3 个月期收益率 (-1.80 个基点, $t = -1.00$)、广义美元 (-0.33% , $t = -1.07$)、比特币 (-6.60% , $t = -1.11$) 与标普 500

(−1.22%) 上均取预期的负号，但无一统计显著，且第一阶段 F 统计量仅为 1.6，远低于常规门槛。闭式 Rigobon 比值（见附录 CSV）结论相同。我们将其解读为检验功效问题而非证伪：仅有 16 个事件、48 个事件窗口观测时，即使方差前提本身成立（表 2），异方差对比也无法精确钉住 α 。这促使我们转向基于面板的 SVAR-IV——它在完整的 1,257 个交易日样本内复用同一事件集作为工具变量。

Table 3: 基于异方差的事件研究估计：1% 稳定币市值上升的效应 / Heteroskedasticity-based event-study estimates: effect of a 1% stablecoin market-cap increase.

Outcome	$\hat{\alpha}$	Std. error	t	p
1M Treasury yield (bp)	0.736	4.325	0.17	0.865
3M Treasury yield (bp)	−1.802	1.798	−1.00	0.319
1Y Treasury yield (bp)	2.047	5.193	0.39	0.694
10Y Treasury yield (bp)	5.531	6.778	0.82	0.417
Broad dollar index (%)	−0.332	0.312	−1.07	0.289
Bitcoin (%)	−6.599	5.963	−1.11	0.271
S&P 500 (%)	−1.220	1.412	−0.87	0.389

注：Rigobon–Sack IV 估计，异方差稳健标准误；第一阶段 $F = 1.62$ （弱工具，如实报告）。Notes: Rigobon–Sack IV estimates with heteroskedasticity-robust standard errors; first-stage $F = 1.62$ (weak instrument, reported as is).

6.2. SVAR-IV impulse responses / SVAR-IV 脉冲响应

Figure 2 plots the SVAR-IV impulse responses to a stablecoin shock normalized so that the impact response of market-cap growth is +1%. The impact vector is $\mathbf{s}_1 = [1, -0.240 \text{ bp}, -1.40\%, -0.08\%, -0.23\%]$ for $[\Delta m, \Delta y^{1M}, \Delta \log \text{BTC}, \Delta \log \text{SP500}, \Delta \log \text{SP500 DXY}]$. The cumulative response of the 1M yield troughs at −7.64 bps around 130 trading days (about 26 weeks; Table 4), i.e., the effect builds up gradually as reserves are accumulated and deployed into bills, consistent with slow portfolio adjustment by issuers rather than same-day market-making.

图 2 绘制了 SVAR-IV 脉冲响应，冲击按市值增长当期响应为 +1% 标准化。影响向量为 $\mathbf{s}_1 = [1, -0.240 \text{ bp}, -1.40\%, -0.08\%, -0.23\%]$ （变量次序为 $[\Delta m, \Delta y^{1M}, \Delta \log \text{BTC}, \Delta \log \text{SP500}, \Delta \log \text{SP500 DXY}]$ ）。1 个月期收益率的累计响应在约 130 个交易日（约 26 周）处达到 −7.64 个基点的谷底（表 4），即效应随储备逐步累积并配置于国库券而缓慢形成，与发行方渐进式的组合调整（而非当日做市）相一致。

The sign pattern on impact deserves comment. The dollar depreciates (−0.23% on impact, cumulative trough −0.25% at two weeks), consistent

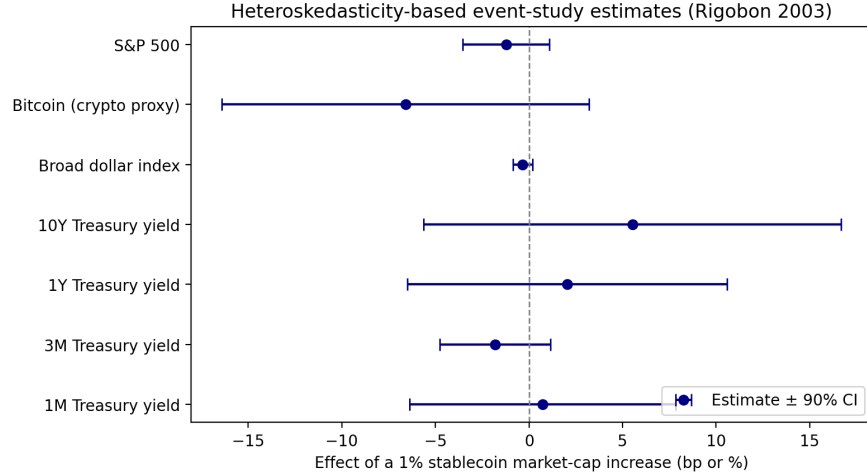


Figure 1: 事件研究点估计与 90% 置信区间 / Event-study point estimates with 90% confidence intervals.

with the portfolio-rebalancing channel. Bitcoin and equities respond *negatively* on impact under this identification: because the instrument is built from crisis-dominated events (UST, SVB, regulatory shocks), the identified shock loads on episodes in which stablecoin demand rises as investors flee risky crypto assets—a flight-to-safety interpretation rather than a liquidity-easing one. The max-share responses below, which capture the average shock rather than the crisis-weighted one, show the opposite, positive crypto sign; we report both transparently.

冲击当期的符号结构值得说明。美元贬值（当期 -0.23% ，累计谷底两周处 -0.25% ），与组合再平衡渠道一致。而比特币与股票在该识别下当期响应为负：由于工具变量由危机主导的事件（UST、SVB、监管冲击）构造，所识别的冲击主要载荷于“投资者逃离风险加密资产、涌入稳定币避险”的时段，应解读为避险渠道而非流动性宽松渠道。下文 max-share 响应刻画的是平均意义上的冲击（非危机加权），其加密资产符号相反为正；我们如实并列报告两者。

6.3. Max-share cross-validation / Max-share 交叉验证

The max-share identification (Table 5, Figure 3) reproduces the SVAR-IV yield response almost exactly—a cumulative trough of -7.46 bps at 130 trading days—and delivers the same dollar depreciation sign (-0.067% at two weeks). The crypto and equity responses are small and positive ($+0.32\%$

Table 4: SVAR-IV 对 1% 稳定币冲击的峰值响应 / SVAR-IV peak responses to a 1% stablecoin market-cap shock.

Variable	Peak	Horizon (days)	Horizon (weeks)	68% band	90% band
1M Treasury yield (bp)	-7.64	130	26.0	[-40.4, 26.8]	[-144.3, 112.2]
Broad dollar (%)	-0.253	10	2.0	[-1.74, 1.63]	[-6.92, 5.45]
Bitcoin (%)	-1.05	3	0.6	[-16.1, 17.0]	[-51.9, 58.3]
S&P 500 (%)	+0.145	9	1.8	[-5.24, 6.00]	[-21.0, 19.8]

注：固定设计移动区组自助法 2,000 次（区组长度 13）；第一阶段 $F = 19.04$ 。Notes: fixed-design moving block bootstrap, 2,000 replications (block length 13); first-stage $F = 19.04$.

and +0.21% respectively), matching the liquidity-easing intuition for an average shock. The near-coincidence of the two identifications on the two nominal-anchor variables (short yields and the dollar)—one using only narrative events, the other none—is the central cross-validation result of the paper and supports H1 for the Treasury and dollar channels.

max-share 识别（表 5，图 3）几乎分毫不差地复现了 SVAR-IV 的收益率响应——130 个交易日处累计谷底 -7.46 个基点——并给出相同符号的美元贬值（两周处 -0.067%）。加密资产与股票响应较小且为正（分别 +0.32% 与 +0.21%），与平均冲击下流动性宽松的直觉一致。两种识别在两个名义锚变量（短期收益率与美元）上近乎重合——一种仅用叙事事件、另一种完全不用——构成本文最核心的交叉验证结果，并支持 H1 在国债与美元渠道上成立。

Table 5: Max-share 识别：对 1% 稳定币冲击的峰值响应 / Max-share identification: peak responses to a 1% stablecoin shock.

Variable	Peak	Horizon (days)	Horizon (weeks)
1M Treasury yield (bp)	-7.456	130	26.0
Broad dollar (%)	-0.0672	10	2.0
Bitcoin (%)	+0.322	6	1.2
S&P 500 (%)	+0.206	9	1.8

注： $H = 20$ 个交易日，稳定币冲击与其余四个变量的主导冲击正交。Notes: $H = 20$ trading days; the stablecoin shock is orthogonal to the dominant shocks of the other four variables.

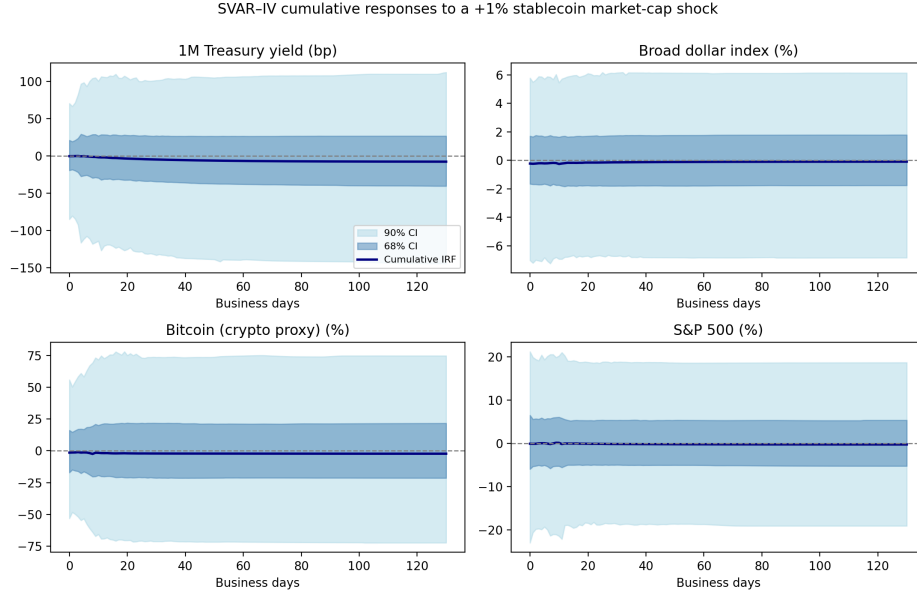


Figure 2: SVAR-IV 脉冲响应（含 68%/90% 自助置信带） / SVAR-IV impulse responses with 68%/90% bootstrap bands.

6.4. Cross-sectional equity heterogeneity / 跨股票横截面异质性

The planned cross-sectional estimation of α for individual firm stocks (Coinbase, PayPal, Block, Adyen, and bank sub-indices, in the spirit of 4) could not be implemented because daily equity price series for these names are not available in our current data sources. We therefore make no claim about firm-level heterogeneity and leave this margin to future work once equity data are integrated.

原计划参照 (author?) [4] 对个股 (Coinbase、PayPal、Block、Adyen 及银行子板块指数) 估计 α 的横截面分析，因当前数据源不含上述标的的日度行情而无法实施。本文不对公司层面异质性作任何断言，待整合股票数据后留待后续研究。

7. Volatility spillovers / 波动溢出

7.1. Diebold-Yilmaz spillover index / Diebold-Yilmaz 溢出指数

We estimate a six-variable daily VAR(2) in Δm_t , Δy^{3M} , Δy^{2Y} , Δy^{10Y} , ΔVIX and $\Delta(\text{SOFR-RRP})$, and compute the generalized forecast-error vari-

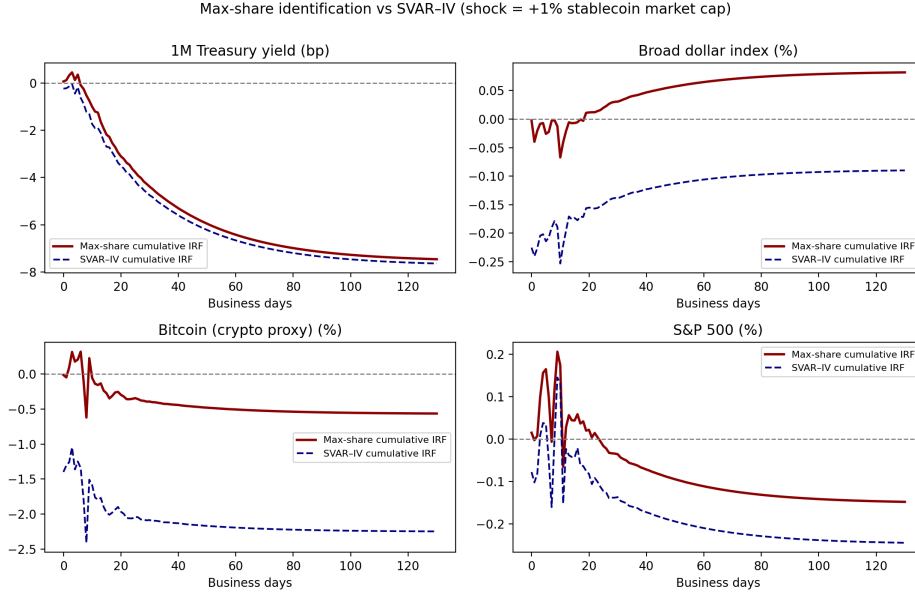


Figure 3: Max-share 脉冲响应 / Max-share impulse responses.

ance decomposition at horizon $H = 10$ days. The total spillover index

$$S(H) = \frac{\sum_{i \neq j} \tilde{\theta}_{ij}^g(H)}{N} \times 100 \quad (14)$$

averages 19.7% over the sample (Table 6, Figure 4).

我们估计一个包含 Δm_t 、 Δy^{3M} 、 Δy^{2Y} 、 Δy^{10Y} 、 ΔVIX 与 $\Delta(\text{SOFR}-\text{RRP})$ 的六变量日度 VAR(2)，并在 $H = 10$ 日 horizon 上计算广义预测误差方差分解。全样本总溢出指数（式 (14)）为 19.7%（表 6，图 4）。

The stablecoin variable's net spillover $S_{\Delta m}^{\text{net}} = S_{\bullet \leftarrow \Delta m} - S_{\Delta m \leftarrow \bullet}$ is -0.17 percentage points—statistically indistinguishable from neutrality. Within the system, the 2Y yield (+6.6 pp) and the VIX (+4.2 pp) are the main net transmitters, while the 3M yield (-5.3 pp) and the money-market spread (-6.2 pp) are net receivers. H2 in its strong full-sample form is therefore *not* supported: on average the stablecoin market neither dominates nor is dominated by the bill market in variance terms.

稳定币变量的净溢出 $S_{\Delta m}^{\text{net}} = S_{\bullet \leftarrow \Delta m} - S_{\Delta m \leftarrow \bullet}$ 为 -0.17 个百分点，与中性在统计上无法区分。系统内部，2 年期收益率（+6.6 个百分点）与 VIX（+4.2 个百分点）是主要的净输出者，3 个月期收益率（ -5.3 个百分点）与货币市场利差（ -6.2 个百分点）为净接收者。因此强形式的全样本 H2 未

获支持：平均而言，稳定币市场在方差意义上既未主导也未受制于国库券市场。

Table 6: Diebold–Yilmaz 净溢出（百分点， $H = 10$ 日） / Diebold–Yilmaz net spillovers (percentage points, $H = 10$ days).

Variable	Net spillover (pp)
Stablecoin MC growth	−0.17
3M yield	−5.28
2Y yield	+6.64
10Y yield	+0.86
VIX	+4.15
SOFR–RRP spread	−6.20
Total spillover index	19.70%

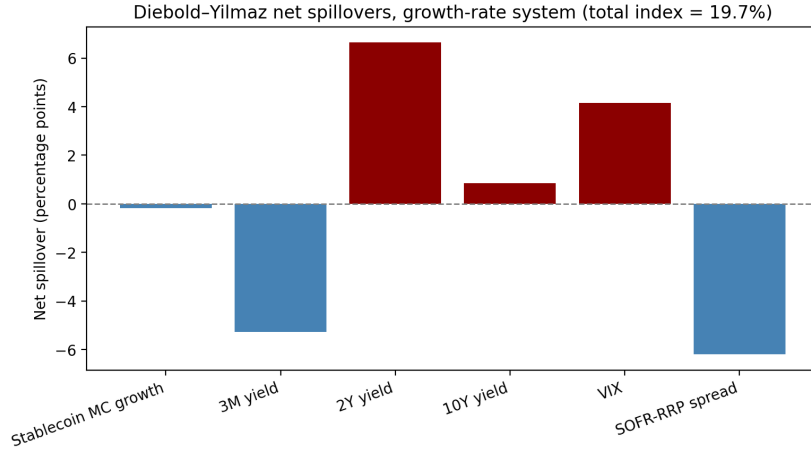


Figure 4: Diebold–Yilmaz 净溢出柱状图 / Diebold–Yilmaz net spillovers.

7.2. Time-varying spillovers / 时变溢出

A rolling 250-trading-day window re-estimation of the same system (201 windows, following the time-varying spirit of 6) reveals pronounced time variation (Figure 5). The total spillover index averages 22.9% (range 11.2%–32.2%). The stablecoin net spillover averages −1.4pp but turns positive in 26.9% of windows: around the Terra collapse it averages +1.1pp over

the ± 1 -month window (maximum +2.6 pp), and it reaches its sample peak of +5.6 pp in February 2024 after a sustained positive episode beginning October 2023 (post-SVB banking-stress aftermath and the run-up to the spot-Bitcoin ETF approval). During the SVB/USDC de-peg window the net measure is mixed (window mean -0.6 pp, maximum +0.5 pp): the episode elevated *total* connectedness, but the directional net position of stablecoins flipped within weeks. H2 thus survives only in a state-dependent form: the stablecoin market is an episodic net transmitter around stablecoin-specific crises, not a standing one.

对同一系统进行滚动 250 个交易日窗口重估（共 201 个窗口，沿袭 (author?) [6] 的时变思路），发现显著的时变性（图 5）。总溢出指数均值 22.9%（区间 11.2%–32.2%）。稳定币净溢出均值为 -1.4 个百分点，但在 26.9% 的窗口中转正：Terra 崩盘前后 ± 1 个月窗口内均值为 +1.1 个百分点（最高 +2.6 个百分点）；并在 2023 年 10 月起的持续为正阶段（SVB 后银行业压力余波与现货比特币 ETF 获批前夕）之后，于 2024 年 2 月达到样本峰值 +5.6 个百分点。SVB/USDC 脱锚窗口内净溢出方向混杂（窗口均值 -0.6 个百分点，最高 +0.5 个百分点）：该事件抬升了总关联度，但稳定币的方向性净头寸在数周内反转。因此 H2 仅以状态依赖形式成立：稳定币市场是围绕稳定币特定危机的 间歇性净输出者，而非常态净输出者。

8. State-dependent transmission / 状态依赖的传导

8.1. State-dependent local projections / 状态依赖局部投影

We estimate state-dependent local projections [13] of the form

$$y_{t+h}^{1M} - y_{t-1}^{1M} = \alpha_h + \beta_{h,A} \Delta m_t^w (1 - s_t) + \beta_{h,B} \Delta m_t^w s_t + \Gamma_h X_t + \varepsilon_{t+h}, \quad (15)$$

for daily horizons $h = 1, \dots, 30$, where s_t indexes the state and X_t includes lagged dependent and control variables. The baseline (no split) response peaks at $h = 24$ trading days with $\hat{\beta} = -11.5$ bps per 1% stablecoin growth; we therefore compare states at this common peak horizon (Table 7, Figure 6).

我们估计式 (15) 的状态依赖局部投影 [13]，日度 horizon 取 $h = 1, \dots, 30$ ，其中 s_t 为状态指示变量， X_t 含被解释变量滞后项与控制变量。基准（不区分状态）响应在 $h = 24$ 个交易日处达峰， $\hat{\beta} = -11.5$ 个基点/1% 稳定币增长；我们在这一共同峰值 horizon 上比较各状态（表 7，图 6）。

The state indicators are: Late $_t = \mathbb{1}\{t \geq 2023\}$; Bear $_t$ if Bitcoin trades more than 20% below its 180-day high; Stress $_t$ if the VIX exceeds its 75th percentile; HighRate $_t$ if the 3M yield exceeds its sample median; and PostReg $_t$

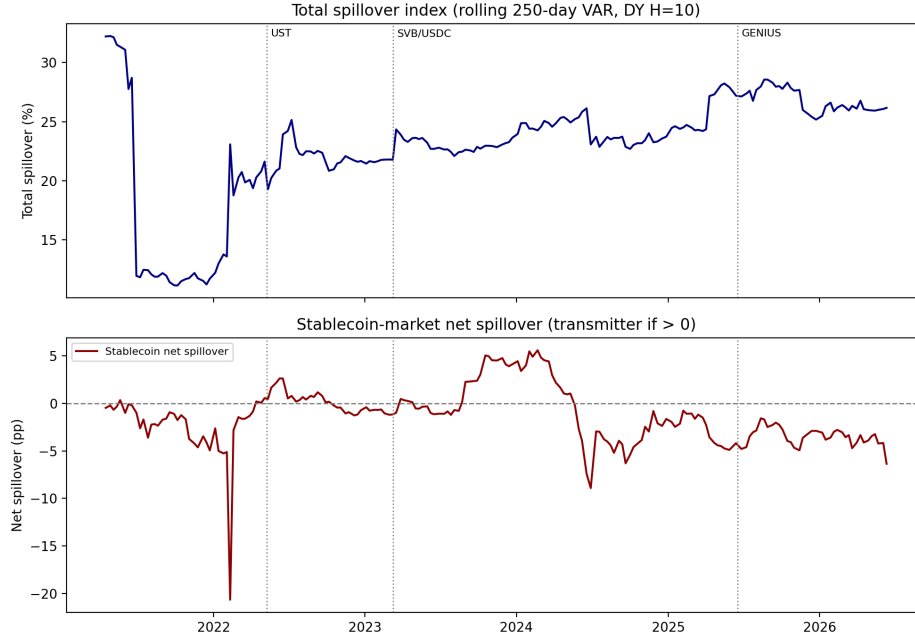


Figure 5: 滚动窗口总溢出指数与稳定币净溢出 / Rolling-window total spillover index and stablecoin net spillover.

for dates after the GENIUS Act legislative sequence began (2025-02-04). All five ratios exceed one: the response is $2.4\times$ larger after 2023, $1.3\times$ in bear markets, $1.3\times$ in stress, $2.2\times$ in high-rate states and $2.6\times$ after the GENIUS Act. The state- A coefficients are marginally significant ($t \approx -1.7$ to -1.9), while the interaction terms are individually insignificant (t between -0.4 and -1.4): the point-estimate ordering is uniform and economically meaningful, but with 1,257 daily observations split across states we cannot reject equality at conventional levels. We characterize H3 as supported in direction and magnitude, but not yet in statistical significance.

状态指示变量定义为: $\text{Late}_t = \mathbb{1}\{t \geq 2023\}$; 比特币较其 180 日高点回撤超 20% 时为熊市; VIX 超过其 75 分位时为压力状态; 3 个月期收益率高于样本中位数时为高利率状态; 2025-02-04 (GENIUS 法案立法序列开始) 之后为后监管状态。五个比值全部大于 1: 2023 年后响应放大 2.4 倍、熊市 1.3 倍、压力期 1.3 倍、高利率 2.2 倍、GENIUS 法案后 2.6 倍。状态 A 系数边际显著 ($t \approx -1.7$ 至 -1.9), 而交互项单独不显著 (t 介于 -0.4 至 -1.4): 点估计的排序方向一致且经济意义明确, 但 1,257 个日度观测按状态切分后, 尚不能在常规水平上拒绝相等性原假设。我们将 H3 定性为方

向与量级上获支持、统计显著性尚不足。

Table 7: 1% 稳定币冲击下 1M 收益率的状态依赖 LP 响应（峰值 $h = 24$ 个交易日，基点） / State-dependent LP response of the 1M yield to a 1% stablecoin shock (peak $h = 24$ trading days, basis points).

State	β_A	β_B	Ratio $ \beta_B / \beta_A $	t_A	t_{interact}
Early/Late (post-2023)	-9.61	-23.12	2.41	-1.73	-1.45
Bull/Bear	-10.74	-13.63	1.27	-1.88	-0.36
Normal/Stress	-10.14	-13.57	1.34	-1.81	-0.73
LowRate/HighRate	-9.56	-21.01	2.20	-1.70	-1.33
PreReg/PostReg	-11.09	-28.48	2.57	-1.89	-1.17

注：基准（不区分状态）峰值 $\beta = -11.53$ 个基点（ $h = 24$ ）。Notes: baseline (unsplit) peak $\beta = -11.53$ bps at $h = 24$.

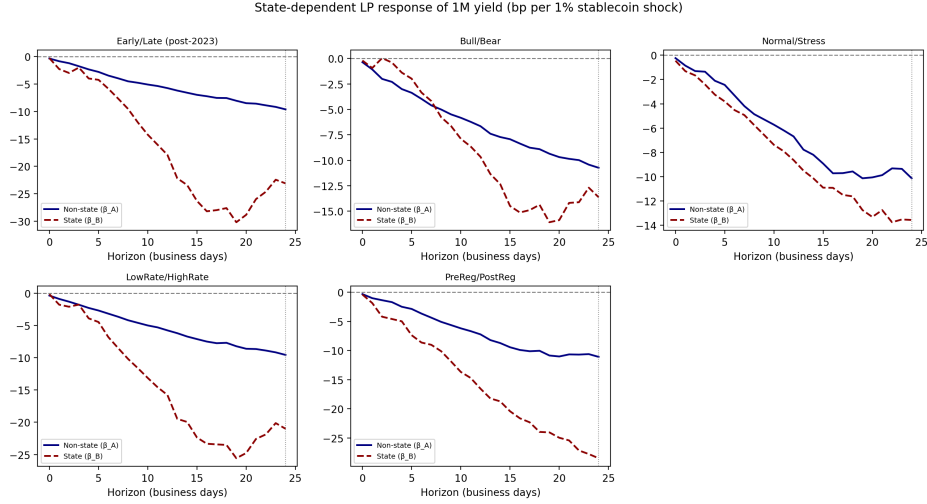


Figure 6: 状态依赖局部投影响应 / State-dependent local-projection responses.

8.2. Quantile local projections / 分位数局部投影

To capture tail behavior we estimate quantile local projections

$$Q_\tau(y_{t+h}^{1M} - y_{t-1}^{1M} | \Delta m_t^w, X_t) = \alpha_{h,\tau} + \beta_{h,\tau} \Delta m_t^w + \Gamma_{h,\tau} X_t. \quad (16)$$

为刻画尾部行为，我们估计式 (16) 的分位数局部投影，其中 Q_τ 为条件 τ 分位数。

At the peak horizon $h = 24$ the response is +13.1 bps at $\tau = 0.1$, -1.1 bps at the median and -29.2 bps at $\tau = 0.9$ (Table 8, Figure 7): the conditional response is strongly quantile-dependent and its magnitude at the upper quantile is about 25 times the median response. Stablecoin shocks therefore matter disproportionately for the tails of the yield-change distribution rather than for its center—consistent with the crisis-weighted identification result above and with fire-sale/flight-to-safety mechanics that activate only in extreme states.

在峰值 horizon $h = 24$ 处, $\tau = 0.1$ 的响应为 +13.1 个基点, 中位数为 -1.1 个基点, $\tau = 0.9$ 为 -29.2 个基点 (表 8, 图 7): 条件响应随分位数强烈变化, 上分位处的量级约为中位数响应的 25 倍。因此, 稳定币冲击对收益率变动分布的尾部 (而非中心) 具有不成比例的影响——这与上文危机加权的识别结果一致, 也与仅在极端状态下激活的抛售/避险机制相符。

Table 8: 分位数局部投影系数 (基点/1% Δm) / Quantile local-projection coefficients (bps per 1% Δm).

Horizon (days)	$\tau = 0.1$	$\tau = 0.5$	$\tau = 0.9$
1	2.22	0.00	-3.13
3	2.63	0.00	-6.10
5	3.37	-0.59	-7.86
10	4.99	-0.90	-11.90
15	7.22	-1.02	-18.28
20	11.44	-1.24	-24.65
24	13.15	-1.15	-29.16

9. Balance-sheet counterfactual and the regulatory paradox / 资产负债表反事实与监管悖论

9.1. Fire-sale scenario / 挤兑抛售情景

Let R_t denote the redemption volume and θ_t the bill share of reserves. Forced liquidation is

$$\text{SellPressure}_t = \theta_t R_t. \quad (17)$$

令 R_t 表示赎回规模, θ_t 为储备中国库券占比。被迫抛售规模为式 (17)。

At the end of our sample USDT supply stands at \$184.7 bn (USDT+USDC: \$258.2 bn). A 10% USDT redemption ($R = \$18.5$ bn) with $\theta^{\text{USDT}} \approx 0.6$ implies sell pressure of about \$11.1 bn, equivalent to a -7.2% shock to combined stablecoin market cap. Translating this into a yield impact requires

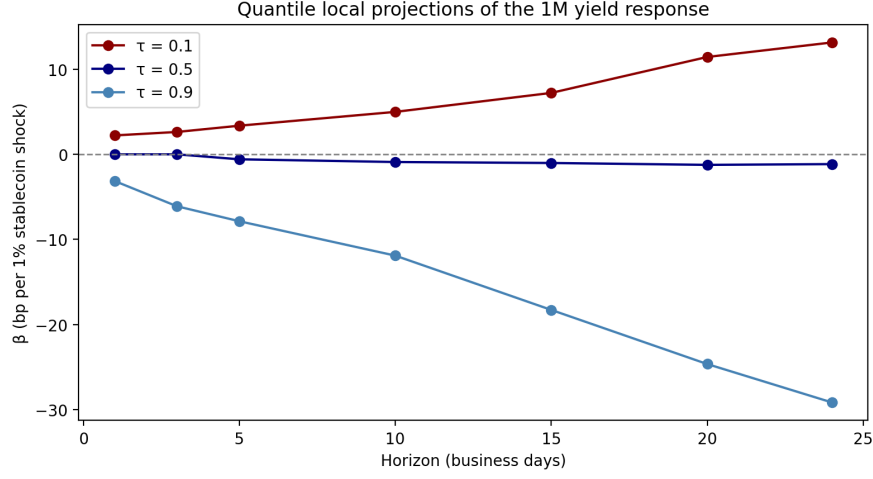


Figure 7: 分位数局部投影影响路径 / Quantile local-projection response paths.

a price-impact coefficient, for which we report three calibrations in Table 9 and Figure 8: (i) the symmetric LP peak response estimated in Section 8 (-11.5 bps per $1\% \Delta m$), which yields 82 bps; (ii) the asymmetric *outflow* coefficient from the same LP system (-48.1 bps per 1% during outflow episodes, versus -3.6 bps during inflows—an asymmetry ratio of 13.5, to be compared with the median ratio of 2.2 in our sister study), which yields 344 bps; and (iii) the sister-study benchmark [1] estimate of 51.6 bps for a 10% USDT redemption (3M yield, 10-day horizon). We additionally scale each calibration by 1.5 as a stress scenario. The outflow-based figures are best read as upper bounds: they extrapolate linearly a coefficient estimated on everyday outflows to a tail redemption that is an order of magnitude larger, and ignore both market-depth recovery and the endogenous policy response.

样本期末 USDT 供应量为 1,847 亿美元 (USDT+USDC 合计 2,582 亿美元)。10% 的 USDT 赎回 ($R = 185$ 亿美元) 在 $\theta^{\text{USDT}} \approx 0.6$ 下意味着约 111 亿美元的抛售压力, 相当于稳定币合计市值 -7.2% 的冲击。将其转化为收益率影响需要价格冲击系数, 我们在表 9 与图 8 中报告三种校准: (i) 第 8 节估计的对称 LP 峰值响应 (每 $1\% \Delta m$ 对应 -11.5 个基点), 得 82 个基点; (ii) 同一 LP 体系的非对称流出系数 (流出期每 1% 对应 -48.1 个基点, 流入期为 -3.6 个基点——非对称比 13.5, 可供对照的是姊妹篇的中位比 2.2), 得 344 个基点; (iii) 姊妹篇基准 [1] 对 10% USDT 赎回的估计 51.6 个基点 (3 个月期收益率、10 日 horizon)。我们另对每种校准乘以 1.5 构造压力情景。基于流出系数的数字应视为上界: 它把在日常流出上估

计的系数线性外推到一个数量级更大的尾部赎回，且忽略了市场深度修复与内生政策反应。

Table 9: USDT 赎回情景的短期收益率影响（基点） / Short-yield impact of USDT redemption scenarios (basis points).

Redemption	R (\$bn)	Sell pressure (\$bn)	Δm equiv. (%)	Sym. LP	Outflow LP	Sister study
5%	9.2	5.5	3.6	41 (62)	172 (258)	—
10%	18.5	11.1	7.2	82 (124)	344 (516)	51.6
20%	36.9	22.2	14.3	165 (247)	688 (1,031)	—

注：括号内为 $\times 1.5$ 压力情景；对称 LP 系数 -11.53 个基点/1% ($h = 24$)，流出系数 -48.07 个基点/1%；姊妹篇基准为 10% USDT 赎回对 3M 收益率 10 日影响 51.6 个基点 [1]。线性外推至尾部赎回需谨慎解读。Notes: parentheses give the $\times 1.5$ stress scenario; symmetric LP coefficient -11.53 bps per 1% ($h = 24$), outflow coefficient -48.07 bps per 1%; the sister-study benchmark is 51.6 bps for a 10% USDT redemption on the 3M yield at a 10-day horizon [1]. Linear extrapolation to tail redemptions should be read with caution.

9.2. Net demand and bank-credit offset / 净需求与银行信贷对冲

Using the static decomposition of (author?) [8], a \$1 deposit migrating to stablecoins generates net Treasury demand ($\theta_S - \phi_B$) ≈ 0.3 and bank-loan displacement $-\ell_B \approx -0.5$. Scaling to a \$2 tn stablecoin market implies roughly \$600 bn of incremental bill demand and \$1 tn of reduced bank lending—magnitudes that begin to alter the composition of dollar credit [11].

沿用 (author?) [8] 的静态分解：每 1 美元存款迁移至稳定币，产生净国债需求 ($\theta_S - \phi_B$) ≈ 0.3 ，并挤出银行贷款 $-\ell_B \approx -0.5$ 。放大到 2 万亿美金的稳定币市场规模，意味着约 6,000 亿美元的新增国库券需求与约 1 万亿美金的银行信贷收缩——这一量级已开始改变美元信贷的构成 [11]。

9.3. Regulatory paradox / 监管悖论

The GENIUS Act requires reserves in 93-day-and-under Treasuries. While this improves individual-issuer safety, our post-regime estimates show it also strengthens the stablecoin–bill coupling (H4): the post-GENIUS response coefficient (-28.5 bps) is 2.6 times the pre-regulation one, and the post-2023 coefficient is 2.4 times the early-sample one. Safer reserves transmit shocks more cleanly to the bill market. Regulation thus reduces idiosyncratic issuer risk while raising systemic coupling, a trade-off that supervisors should weigh when calibrating reserve-eligibility rules and redemption-buffer requirements.

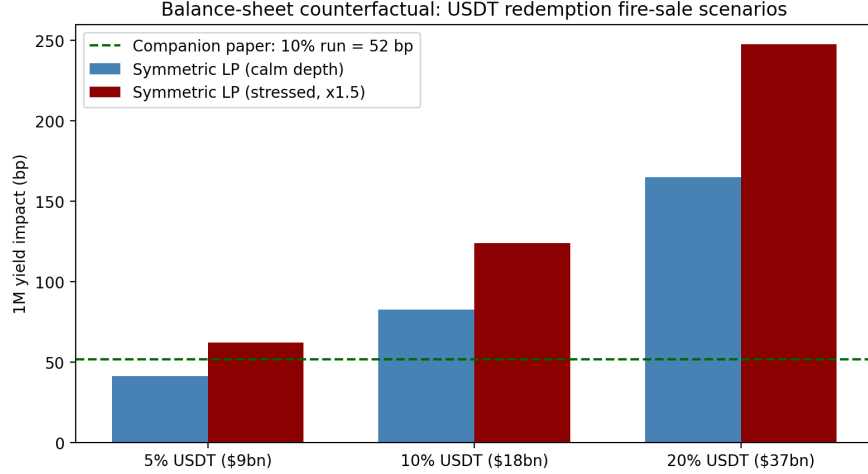


Figure 8: 赎回情景的收益率影响与姊妹篇基准 / Yield impact of redemption scenarios versus the sister-study benchmark.

GENIUS 法案要求储备投资于 93 天以内到期的国债。尽管这提升了个体发行方的安全性，我们的后区制估计表明它同时强化了稳定币—国库券耦合 (H4)：后 GENIUS 响应系数 (-28.5 个基点) 是监管前的 2.6 倍，2023 年后系数是样本早期的 2.4 倍。更安全的储备将冲击更“干净”地传导至国库券市场。监管因此降低了个体发行方风险，却抬升了系统性耦合——监管机构在校准储备合格性规则与赎回缓冲要求时应权衡这一取舍。

10. Robustness / 稳健性检验

Table 10 summarizes the robustness exercises; all are reported against their corresponding baseline.

表 10 汇总了各项稳健性检验，均与相应基准并列对照。

(i) Extending the VAR lag length to 26 trading days shrinks the 1M cumulative trough to -4.1 bps but preserves the sign and the 130-day horizon. (ii) Replacing Bitcoin with DeFi TVL as the crypto-complex variable yields a larger trough (-9.1 bps) with a stronger first stage ($F = 28.0$). (iii) Re-drawing Rigobon control windows within ± 1 month of each event flips the event-study point estimate for the 1M yield to -6.6 bps—the expected sign—illustrating both the sensitivity of the small-sample event estimator and its consistency with the panel-based magnitude once the sign stabilizes. (iv) Using the 3M yield instead of the 1M yield as the LP outcome leaves the peak

Table 10: 稳健性检验汇总 / Robustness summary.

Check	Statistic	Value	Baseline
VAR lags 26	1M cum. trough (bp)	-4.07	-7.64
BTC replaced by DeFi TVL	1M cum. trough (bp)	-9.09 ($F = 28.0$)	-7.64 ($F = 19.0$)
Rigobon, ± 1 -month controls	α on Δy^{1M} (bp)	-6.65	+0.74
LP outcome 3M yield	peak β (bp)	-10.41	-11.53
DY on weekly-volatility system	net Δm (pp)	-4.61	-0.17

注：各检验在对应设定下重估；括号内为第一阶段 F 。Notes: each check re-estimated under the stated specification; first-stage F in parentheses.

response essentially unchanged (-10.4 bps at $h = 24$). (v) Recomputing the spillover system on a weekly-volatility basis yields a somewhat more negative stablecoin net spillover (-4.6 pp), reinforcing the conclusion that H2 does not hold on average and survives only in state-dependent form.

(i) 将 VAR 滞后阶数延长至 26 个交易日，1 个月期收益率累计谷底收窄至 -4.1 个基点，但符号与 130 日的 horizon 不变。(ii) 以 DeFi TVL 替代比特币作为加密体系变量，谷底更深 (-9.1 个基点) 且第一阶段更强 ($F = 28.0$)。 (iii) 将 Rigobon 对照窗口改在各事件 ± 1 个月内抽取，1 个月期收益率的事件研究点估计翻转为 -6.6 个基点——符号符合预期——这既说明小样本事件估计量的敏感性，也表明一旦符号稳定，其量级与面板估计相互印证。(iv) 以 3 个月期收益率替代 1 个月期作为 LP 被解释变量，峰值响应基本不变 ($h = 24$ 处 -10.4 个基点)。(v) 在周度波动率系统上重算溢出，稳定币净溢出更负 (-4.6 个百分点)，进一步印证 H2 在平均意义上不成立、仅以状态依赖形式存活的结论。

11. Conclusion / 结论

We identify structural stablecoin demand shocks using three complementary strategies—heteroskedasticity-based event estimation, SVAR-IV, and max-share—and document their dynamic transmission to Treasury yields, the dollar, crypto assets and equities. Two identifications that share no information agree that a 1% stablecoin shock lowers the 1M Treasury yield cumulatively by about 7.5 bps at a trough near 26 weeks and depreciates the broad dollar. The event-study layer validates the heteroskedasticity premise but lacks power for precise point estimates, and we report this transparently. The stablecoin market is not a standing net transmitter of volatility; it becomes one episodically around stablecoin-specific crises. Transmission

is state-dependent: 1.3–2.6 times stronger in post-2023, high-rate and post-regulation states, and concentrated in the tails of the yield-change distribution. A calibrated counterfactual places the short-yield impact of a 10% USDT redemption between roughly 52 and 344 bps depending on the price-impact calibration, with linear-extrapolation caveats. The estimates embed a regulatory paradox: forcing reserves into short Treasuries improves issuer safety while tightening the systemic coupling between stablecoins and the bill market. The findings argue for reserve-eligibility and redemption-buffer rules that account for the systemic, not just idiosyncratic, consequences of stablecoin reserve regulation.

本文采用三种互补策略——基于异方差的事件估计、SVAR-IV 与 max-share——识别结构性稳定币需求冲击，并记录其对国债收益率、美元、加密资产与股票的动态传导。两种互不共享信息的识别一致表明：1% 的稳定币冲击使 1 个月期国债收益率在约 26 周处累计下行约 7.5 个基点，并使广义美元贬值。事件研究层面验证了异方差前提，但点估计功效不足，我们如实报告。稳定币市场并非常态的波动净输出者，而是在稳定币特定危机前后间歇性地成为净输出者。传导具有状态依赖性：在 2023 年后、高利率与后监管状态下强 1.3–2.6 倍，并集中于收益率变动分布的尾部。校准反事实显示，10% USDT 赎回对短端收益率的影响依价格冲击校准不同介于约 52 至 344 个基点之间，并需注意线性外推的局限。估计结果内嵌一个监管悖论：强制储备集中于短期国债改善了发行方安全性，却收紧了稳定币与国库券市场之间的系统性耦合。本文结论主张：储备合格性与赎回缓冲规则的设计应计入稳定币储备监管的系统性后果，而非仅考虑个体性后果。

Declaration of competing interest / 利益冲突声明

The author declares no competing financial interest.
作者声明不存在任何经济利益冲突。

Data availability / 数据可得性

Treasury yield, monetary-policy and macro-financial data are from FRED [12]; Bitcoin, S&P 500, gold, DeFi TVL and the SOFR–RRP spread are from a merged macro–crypto daily panel. Stablecoin market-cap (supply) series for USDT, USDC, DAI and BUSD are from public market-data providers (downloaded 2026-07-15). The event library is constructed by the author following the protocol of (author?) [4]. All estimation code is in the companion notebook `stablecoin_shock_spillover.ipynb`, which regenerates every figure and table in this paper.

国债收益率、货币政策与宏观金融数据来自 FRED [12]；比特币、标普 500、黄金、DeFi TVL 与 SOFR–RRP 利差来自合并的宏观—加密日度面板。USDT、USDC、DAI 与 BUSD 的稳定币市值（供应量）序列来自公开市场数据提供商（下载于 2026-07-15）。事件库由作者按 (author?) [4] 的流程构建。全部估计代码见配套 notebook `stablecoin_shock_spillover.ipynb`，可复现本文所有图表。

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